

ASTRONOMICAL OBSERVATORIES

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SINCE ABOUT THE early 20th century, many new astronomical observatories have been built, housing ever-larger telescopes. Many of these instruments are visible-light telescopes, but with continuing technological advances, telescopes for studying other parts of the electromagnetic spectrum have also been built, such as radio telescopes and gamma-ray telescopes.

OBSERVATORY TELESCOPES

Most observatory telescopes are placed away from the air and light pollution of urban areas and at high altitude to minimize atmospheric distortion. The size of a telescope is also important: the larger a telescope's aperture, the greater its light-gathering power. Objective lenses for refractors cannot be made more than about 40 in (1 m) across—the size of the Yerkes refractor (below left)—but single-piece mirrors can be made up to about 26.25 ft (8 m) across—the size of the Gemini South telescope (right). Using segmented mirrors, reflectors can be made even larger. For example, the Gran Telescopio Canarias has a segmented mirror 34 ft (10.4 m) across.



GEMINI SOUTH TELESCOPE

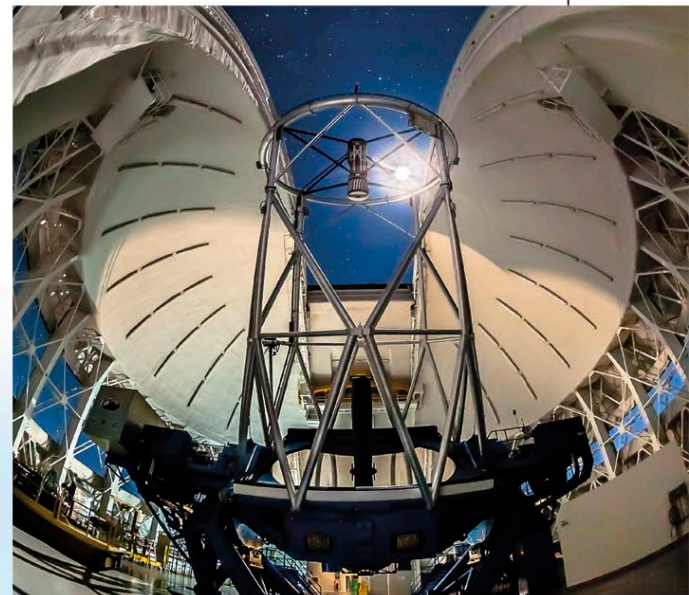
Moonlight casts shadows inside the dome of the Gemini South telescope, which sits at an altitude of 8,980 ft (2,737 m) on Cerro Pachón, a mountain in the Chilean Andes. An identical telescope, Gemini North, is situated near the summit of Mauna Kea on the island of Hawaii.

YERKES REFRACTOR

Refracting telescopes reached their pinnacle with the 40 in (1 m) instrument at Yerkes Observatory, Wisconsin, shown on the left. Opened in 1897, it remains the largest refractor ever built.

PALOMAR OBSERVATORY

Like all large, modern observatories, the Palomar in California, was built at high altitude (5,617 ft/1,712 m) for optimum viewing conditions.



PARANAL OBSERVATORY

Situated at an altitude of 8,645 ft (2,635 m) on Cerro Paranal in northern Chile, the Very Large Telescope (VLT) is one of the largest modern telescope arrays, consisting of four 26.9 ft (8.2 m) reflectors. The telescopes operate at visible light and infrared wavelengths and can be used either independently or in combination for greater resolution.